

Project Proposal

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Project Title

Numerical Root-Finding Methods Library in Python

Description

Build a Python library implementing three root-finding methods — Bisection, Newton-Raphson, and Secant. The project will use Git for version control, pytest for testing, a Makefile for automation, and matplotlib for convergence plots.

What Will Be Done

- Implement Bisection, Newton-Raphson, and Secant methods
- Write unit tests using pytest
- Set up Git repository with proper commits
- Automate tasks using a Makefile
- Plot and compare convergence of all three methods

Timeline

Week 1	Implement all three methods + unit tests
Week 2	Git setup, Makefile, convergence plots
Week 3	Final report and submission